PAPER NAME

ASML Company Analysis.docx

WORD COUNT	CHARACTER COUNT
1094 Words	**6881 Characters**
PAGE COUNT	FILE SIZE
3 Pages	**23.2KB**
SUBMISSION DATE	REPORT DATE
Feb 12, 2026 9:52 AM GMT+5:30	**Feb 12, 2026 9:52 AM GMT+5:30**

7% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

- 4% Internet database
- 0% Publications database
- Crossref database
- Crossref Posted Content database
- 7% Submitted Works database

ASML: Advanced Semiconductor Materials Lithography

Executive Summary

ASML is the single most critical but invisible component of the world's infrastructure of modern technology. ASML, being the single supplier of its extreme ultraviolet lithography equipment, the most intricate production equipment the world has to offer, calls the shots on the gate through which what may be the most advanced semiconductors to date are being manufactured. The pricing of the machines will fall in the range of $200-380 million, which will result in revenues of '2024 of €28.3 billion. This will give the company a net income of €7.6 billion. The EUV monopoly of the company, developed with decades of investment in R&D and strategic alliances, has given the company a technological barrier that no competitor has been able to cross. For the year 2025, the company is looking at revenues of €30-35 billion due to the surge in demand for semiconductors due to AI and the rollout of High-NA EUV.

Company Background & History

ASML Holding NV, which, although being based in the Netherlands, has its operations based in the city of Veldhoven, has grown from being a regional player to becoming a global technology cornerstone via its strategy of building ecosystems. Instead of engaging in production, it makes use of its 5,150+ suppliers to procure 85% of its component production. This, along with investing ~14% every year in R&D ($3.5+ billion), has helped create an innovation engine.

Historical Evolution: ASML pioneered Deep Ultraviolet (DUV) lithography through the 1990s-2000s, but recognized a fundamental limitation—as semiconductor nodes shrank below 7nm, DUV became economically unsustainable. The company has invested 6 billion Euros and twenty years into EUV. This included facing never-before-experienced challenges in light, lenses, vacuums, and metrology. The early 2010s saw ASML introduce the first commercially viable EUV systems into the market. Now, High NA EUV systems, doubling the resolution of the pattern, are going into production. This is another inflection point for the company. The first customers of the step and repeat systems had invested up to $380 million.

Target Market, Competitive Position & What Problem They Solve

Target Market: ASML has Tier-1 foundry customers such as TSMC, Samsung, and Intel, who need to make leading-edge chipsets. In 2024, 40% of sales came from China, but export restrictions are limiting technology export. The TAM for EUV-related equipment hovers between $8 to $12 billion.

The Physics Challenge: ASML addressed the key question of how to print features smaller than the wavelength of light itself using light. With the ability to use 13.5 nm wavelengths, the consequence has been the ability to print 3 nm, 2 nm nodes, and eventually patterns on the way to 1 nm without incurring the expense required in bulk patterning techniques. High-NA EUV lithography offers even more precise dimensions than current technology, continuing Moore's Law into the late 2020s.

Customer Pain Points: Chipmakers face the problems of scaling margins, capex efficiency, time-to-market constraints, especially to AI, and bad yields at ultraviolet dimensions. ASML offers more-throughput, increased accuracy (overlay accuracy less than 1.5 nm) solutions and is more mature than competitor solutions.

Competitive Landscape: ASML does not have 100% market share in EUV by accident, but by a carefully crafted moat: (1) Technical dominance in expertise in the field of optics, photonics, vacuum systems and software; (2) Supply chain dominance in the development of switching costs; (3) Installed base in recurring revenue; (4) Geopolitical factors in the protection of its position.

Business Model, Risks & Recommendations

Business Model: ASML generates revenue through new sales system ($200-380M per unit, 50%+ gross margins), which has installed a base management (spare parts, upgrades at 60%+ margins, ~25-30% of quarterly revenue), used system sales, and emerging software/data services. Outsourced components (85% of value) drive exceptional leverage—gross margins hover at 52-53%, enabling simultaneous R&D investment and shareholder returns while maintaining 25-28% net margins.

Key Risks:
Geopolitical/Export Controls: China 40% of our sales, is subject to some restrictions, additional restrictions have some potential to remove significant revenues
Industry Cyclicality: Semiconductor Capex is a great cyclical phenomenon, so it produces revenue volatility, and a 20-30% downturn may signify 100%
High-NA Execution Risk Technical risks related to pellicles, stochasticity in photoresist and accuracy in overlay can make it difficult to be profitable
Supply Chain Vulnerabilities In the case of a large supply chain - 5,150+ suppliers, risks
Competitive Disruption: Although it's a remote prospect China or next generation lithography could crack the monopoly one day
Tariff Uncertainty: A rising level of protectionism is impacting the planning of buyer Capex
Strategic Recommendations:
Accelerate High-NA Ecosystem: Joint Development with Customers (TSMC, Samsung, Intel) for pellicles, photoresists, process recipes; Have dedicated support teams
Geopolitical Strategy: Establish regional service centres, Multi-tier product services, and a better governance of relationship as a means of overcoming export restrictions.
Adjacent Market Expansions In more, the semiconductor packaging/chiplet assembly equipment; $2-$5B addressable market, leverage in existent precision expertise
Next-Gen R&D: Safe from long-term disruption by investing in Extreme Deep Ultraviolet at 6.7nm and other alternative lithography concepts.
AI-driven software develops machine learning for real-time prediction of overlays, predictive maintenance and optimizing the process. Revenue recurring: $500M-1B.
Talent Investment: Develop career paths and innovation incentives, ensure University partnerships on assistance in maintaining technological edge.
Strategic M&A: Acquire complementary metrology, material or software capabilities to accelerate the growth of an ecosystem

Conclusion

ASML is said to be an excellent example of dominance by strategy, where monopoly power is achieved through a technological differentiation. Its EUV systems are virtually unmatched in state-of-the-art semiconductor production, driving high profitability on the basis of inelastic demand. Yet, such dominance is attained in a dynamic geopolitical landscape and technological environment. Export restrictions and business cycle impacts, High-NA technical challenges and continuous threats of disruption simply need to be considered. The foregoing suggestions to address these threats, which were all made, above, to accelerate high NA maturation, address geopolitical considerations, grow revenue channels, and maintain technological leadership, are all meant to facilitate ASML's continued dominance through the coming years. Ultimately, the success of ASML is due to ASML being the company that solves the most critical semiconductor challenges and prepares to solve the ones to come.

References

ASML Holding NV. (2025, July 15). Q2 2025 Financial Results. https://www.asml.com/en/news/press-releases/2025/q2-2025-financial-results

Euronews Business. (2025, January 29). ASML reports boost in full-year sales as it shares outlook for 2025. https://www.euronews.com/business/2025/01/29/asml-reports-boost-in-full-year-sales-as-it-shares-outlook-for-2025

TrendForce. (2025, August 31). ASML EUV Dominance & China's Semiconductor Path. https://www.trendforce.com/insights/asml-euv

AIME by AInvest. (2025, January 7). What are the risks associated with investing in ASML? https://www.ainvest.com/aime/share/what-are-the-risks-associated-with-investing-in-asml-bf156f/

McKinsey Electronics. (2025, August 6). EUV Lithography Issues Engineers Face. https://www.mckinsey-electronics.com/post/10-hidden-challenges-of-extreme-ultraviolet-euv-lithography-that-no-one-talks-about

7% Overall Similarity

Top sources found in the following databases:

- 4% Internet database
- 0% Publications database
- Crossref database
- Crossref Posted Content database
- 7% Submitted Works database

TOP SOURCES

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	**Hogeschool Rotterdam on 2025-04-07** Submitted works	3%
2	**University of Sussex on 2025-12-20** Submitted works	2%
3	**Tias Business School on 2026-01-25** Submitted works	1%
4	**University College London on 2026-01-05** Submitted works	<1%
5	**Tias Business School on 2026-01-25** Submitted works	<1%

ASML Company Analysis.docx

Fr. Conceicao Rodrigues College Of Engineering

Document Details

Submission ID

trn:oid:::12001:128067733

Submission Date

Feb 12, 2026, 9:52 AM GMT+5:30

Download Date

Feb 12, 2026, 10:02 AM GMT+5:30

File Name

ASML Company Analysis.docx

File Size

23.2 KB

3 Pages

1,094 Words

6,881 Characters

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

ASML: Advanced Semiconductor Materials Lithography

Executive Summary

ASML is the single most critical but invisible component of the world's infrastructure of modern technology. ASML, being the single supplier of its extreme ultraviolet lithography equipment, the most intricate production equipment the world has to offer, calls the shots on the gate through which what may be the most advanced semiconductors to date are being manufactured. The pricing of the machines will fall in the range of $200-380 million, which will result in revenues of '2024 of €28.3 billion. This will give the company a net income of €7.6 billion. The EUV monopoly of the company, developed with decades of investment in R&D and strategic alliances, has given the company a technological barrier that no competitor has been able to cross. For the year 2025, the company is looking at revenues of €30-35 billion due to the surge in demand for semiconductors due to AI and the rollout of High-NA EUV.

Company Background & History

ASML Holding NV, which, although being based in the Netherlands, has its operations based in the city of Veldhoven, has grown from being a regional player to becoming a global technology cornerstone via its strategy of building ecosystems. Instead of engaging in production, it makes use of its 5,150+ suppliers to procure 85% of its component production. This, along with investing ~14% every year in R&D ($3.5+ billion), has helped create an innovation engine.

Historical Evolution: ASML pioneered Deep Ultraviolet (DUV) lithography through the 1990s-2000s, but recognized a fundamental limitation—as semiconductor nodes shrank below 7nm, DUV became economically unsustainable. The company has invested 6 billion Euros and twenty years into EUV. This included facing never-before-experienced challenges in light, lenses, vacuums, and metrology. The early 2010s saw ASML introduce the first commercially viable EUV systems into the market. Now, High NA EUV systems, doubling the resolution of the pattern, are going into production. This is another inflection point for the company. The first customers of the step and repeat systems had invested up to $380 million.

Target Market, Competitive Position & What Problem They Solve

Target Market: ASML has Tier-1 foundry customers such as TSMC, Samsung, and Intel, who need to make leading-edge chipsets. In 2024, 40% of sales came from China, but export restrictions are limiting technology export. The TAM for EUV-related equipment hovers between $8 to $12 billion.

The Physics Challenge: ASML addressed the key question of how to print features smaller than the wavelength of light itself using light. With the ability to use 13.5 nm wavelengths, the consequence has been the ability to print 3 nm, 2 nm nodes, and eventually patterns on the way to 1 nm without incurring the expense required in bulk patterning techniques. High-NA EUV lithography offers even more precise dimensions than current technology, continuing Moore's Law into the late 2020s.

Customer Pain Points: Chipmakers face the problems of scaling margins, capex efficiency, time-to-market constraints, especially to AI, and bad yields at ultraviolet dimensions. ASML offers more-throughput, increased accuracy (overlay accuracy less than 1.5 nm) solutions and is more mature than competitor solutions.

Competitive Landscape: ASML does not have 100% market share in EUV by accident, but by a carefully crafted moat: (1) Technical dominance in expertise in the field of optics, photonics, vacuum systems and software; (2) Supply chain dominance in the development of switching costs; (3) Installed base in recurring revenue; (4) Geopolitical factors in the protection of its position.

Business Model, Risks & Recommendations

Business Model: ASML generates revenue through new sales system ($200-380M per unit, 50%+ gross margins), which has installed a base management (spare parts, upgrades at 60%+ margins, ~25-30% of quarterly revenue), used system sales, and emerging software/data services. Outsourced components (85% of value) drive exceptional leverage—gross margins hover at 52-53%, enabling simultaneous R&D investment and shareholder returns while maintaining 25-28% net margins.

Key Risks:
Geopolitical/Export Controls: China 40% of our sales, is subject to some restrictions, additional restrictions have some potential to remove significant revenues
Industry Cyclicality: Semiconductor Capex is a great cyclical phenomenon, so it produces revenue volatility, and a 20-30% downturn may signify 100%
High-NA Execution Risk Technical risks related to pellicles, stochasticity in photoresist and accuracy in overlay can make it difficult to be profitable
Supply Chain Vulnerabilities In the case of a large supply chain - 5,150+ suppliers, risks
Competitive Disruption: Although it's a remote prospect China or next generation lithography could crack the monopoly one day
Tariff Uncertainty: A rising level of protectionism is impacting the planning of buyer Capex
Strategic Recommendations:
Accelerate High-NA Ecosystem: Joint Development with Customers (TSMC, Samsung, Intel) for pellicles, photoresists, process recipes; Have dedicated support teams
Geopolitical Strategy: Establish regional service centres, Multi-tier product services, and a better governance of relationship as a means of overcoming export restrictions.
Adjacent Market Expansions In more, the semiconductor packaging/chiplet assembly equipment; $2-$5B addressable market, leverage in existent precision expertise
Next-Gen R&D: Safe from long-term disruption by investing in Extreme Deep Ultraviolet at 6.7nm and other alternative lithography concepts.
AI-driven software develops machine learning for real-time prediction of overlays, predictive maintenance and optimizing the process. Revenue recurring: $500M-1B.
Talent Investment: Develop career paths and innovation incentives, ensure University partnerships on assistance in maintaining technological edge.
Strategic M&A: Acquire complementary metrology, material or software capabilities to accelerate the growth of an ecosystem

Conclusion

ASML is said to be an excellent example of dominance by strategy, where monopoly power is achieved through a technological differentiation. Its EUV systems are virtually unmatched in state-of-the-art semiconductor production, driving high profitability on the basis of inelastic demand. Yet, such dominance is attained in a dynamic geopolitical landscape and technological environment. Export restrictions and business cycle impacts, High-NA technical challenges and continuous threats of disruption simply need to be considered. The foregoing suggestions to address these threats, which were all made, above, to accelerate high NA maturation, address geopolitical considerations, grow revenue channels, and maintain technological leadership, are all meant to facilitate ASML's continued dominance through the coming years. Ultimately, the success of ASML is due to ASML being the company that solves the most critical semiconductor challenges and prepares to solve the ones to come.

References

ASML Holding NV. (2025, July 15). Q2 2025 Financial Results. https://www.asml.com/en/news/press-releases/2025/q2-2025-financial-results

Euronews Business. (2025, January 29). ASML reports boost in full-year sales as it shares outlook for 2025. https://www.euronews.com/business/2025/01/29/asml-reports-boost-in-full-year-sales-as-it-shares-outlook-for-2025

TrendForce. (2025, August 31). ASML EUV Dominance & China's Semiconductor Path. https://www.trendforce.com/insights/asml-euv

AIME by AInvest. (2025, January 7). What are the risks associated with investing in ASML? https://www.ainvest.com/aime/share/what-are-the-risks-associated-with-investing-in-asml-bf156f/

McKinsey Electronics. (2025, August 6). EUV Lithography Issues Engineers Face. https://www.mckinsey-electronics.com/post/10-hidden-challenges-of-extreme-ultraviolet-euv-lithography-that-no-one-talks-about